

PREVIOUS YEARS' QUESTIONS

- 41.** What is the value of k that $(2x - 1)$ may be a factor of $4x^4 - (k - 1)x^3 + kx^2 - 6x + 1$? **2012 I**
 (a) 8 (b) 9 (c) 12 (d) 13
- 42.** $x^4 + xy^3 + x^3y + xz^3 + y^4 + yz^3$ is divisible by **2012 I**
 (a) Only $(x - y)$
 (b) Only $(x^3 + y^3 + z^3)$
 (c) Both $(x + y)$ and $(x^3 + y^3 + z^3)$
 (d) None of the above
- 43.** If $x^3 + 5x^2 + 10k$ leaves remainder $-2x$ when divided by $x^2 + 2$, then what is the value of k ? **2012 I**
 (a) -2 (b) -1 (c) 1 (d) 2
- 44.** If $x + \frac{1}{x} = a$, then what is the value of $x^3 + x^2 + \frac{1}{x^3} + \frac{1}{x^2}$? **2012 I**
 (a) $a^3 + a^2$ (b) $a^3 + a^2 - 5a$
 (c) $a^3 + a^2 - 3a - 2$ (d) $a^3 + a^2 - 4a - 2$
- 45.** If $p(x)$ is a common multiple of degree 6 of the polynomials $f(x) = x^3 + x^2 - x - 1$ and $g(x) = x^3 - x^2 + x - 1$, then which one of the following is correct? **2012 I**
 (a) $\rho(x) = (x - 1)^2 (x + 1)^2 (x^2 + 1)$
 (b) $\rho(x) = (x - 1) (x + 1) (x^2 + 1)^2$
 (c) $\rho(x) = (x - 1)^3 (x + 1) (x^2 + 1)$
 (d) $\rho(x) = (x - 1)^2 (x^4 + 1)$
- 46.** What is the value of k which will make the expression $4k^2 + 12k + k$ a perfect square? **2012 II**
 (a) 5 (b) 7 (c) 8 (d) 9
- 47.** The factors of $5px - 10qy + 2rpx - 4qry$ is/are **2013 I**
 (a) Only $(5 + 2r)$
 (b) Only $(px - 2qy)$
 (c) Both $(5 + 2r)$ and $(px - 2qy)$
 (d) Neither $(5 + 2r)$ nor $(px - 2qy)$
- 48.** $(a + 1)^4 - a^4$ is divisible by **2013 I**
 (a) $-2a^2 + 2a - 1$ (b) $2a^3 - 2a - 1$
 (c) $2a^3 - 2a + 1$ (d) $2a^2 + 2a + 1$
- 49.** One of the factors of the polynomial $x^4 - 7x^3 + 5x^2 - 6x + 81$ is **2013 I**
 (a) $x + 2$ (b) $x - 2$ (c) $x + 3$ (d) $x - 3$
- 50.** If the expression $x^3 + 3x^2 + 4x + k$ has a factor $x + 5$, then what is the value of k ? **2013 II**
 (a) -70 (b) 70 (c) 48 (d) -48
- 51.** The quantity which must be added to $(1 - x)(1 + x^2)$ to obtain x^3 is **2013 II**
 (a) $2x^3 + 3x^2 + x + 1$ (b) $2x^3 + x^2 + x - 1$
 (c) $2x^3 - x^2 + x - 1$ (d) $-x^3 + x - 1$
- 52.** $x(y^2 - z^2) + y(z^2 - x^2) + z(x^2 - y^2)$ is divisible by **2013 II**
 (a) Only $(y - z)$
 (b) Only $(z - x)$
 (c) Both $(y - z)$ and $(z - x)$
 (d) Neither $(y - z)$ nor $(z - x)$
- 53.** Consider the following statements
 I. $x + 3$ is the factor of $x^3 + 2x^2 + 3x + 8$.
 II. $x - 2$ is the factor of $x^3 + 2x^2 + 3x + 8$.
 Which of the statement(s) given above is/are correct? **2013 II**
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 54.** $x^3 + 6x^2 + 11x + 6$ is divisible by **2014 I**
 (a) Only $(x + 1)$ (b) Only $(x + 2)$
 (c) Only $(x + 3)$ (d) All of these
- 55.** What should be added to the $x(x + a)(x + 2a)(x + 3a)$, so that the sum be a perfect square?
 (a) $9a^2$ (b) $4a^2$ **2014 I**
 (c) a^4 (d) None of these
- 56.** If $\left(x^2 + \frac{1}{x^2}\right) = \frac{17}{4}$, then what is $\left(x^3 - \frac{1}{x^3}\right)$ equal to? **2014 I**
 (a) $\frac{75}{16}$ (b) $\frac{63}{8}$ (c) $\frac{95}{8}$ (d) None of these
- 57.** What is the remainder when $x^5 - 5x^2 + 125$ is divided by $x + 5$? **2014 II**
 (a) 0 (b) 125 (c) -3125 (d) 3125
- 58.** If $ax + by - 2 = 0$ and $axby = 1$, where $a \neq 0, b \neq 0$, then what is $(a^2x + b^2y)$ equal to? **2014 II**
 (a) $a + b$ (b) $2ab$ (c) $a^3 + b^3$ (d) $a^4 + b^4$
- 59.** If $(x + k)$ is the common factor of $x^2 + ax + b$ and $x^2 + cx + d$, then what is k equal to? **2014 II**
 (a) $(d - b)/(c - a)$ (b) $(d - b)/(a - c)$
 (c) $(d + b)/(c + a)$ (d) $(d - b)/(c + a)$
- 60.** Consider the following statements:
 I. $(a - b - c)$ is one of the factors of $3abc + b^3 + c^3 - a^3$.
 II. $(b + c - 1)$ is one of the factors of $3bc + b^3 + c^3 - 1$.
 Which of the statement(s) given above is/are correct? **2014 II**
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 61.** For what value of k is $(x - 5)$ a factor of $x^3 - 3x^2 + kx - 10$? **2015 I**
 (a) -8 (b) 4 (c) 2 (d) 1
- 62.** If $x + y + z = 0$, then $x^3 + y^3 + z^3 + 3xyz$ is equal to **2015 I**
 (a) 0 (b) $6xyz$ (c) $12xyz$ (d) xyz

- 63.** The expression $x^3q^2 - x^3pt + 4x^2pt - 4x^2q^2 + 3xq^2 - 3xpt$ is divisible by ☑ 2015 I
 (a) Only $(x-1)$ (b) Only $(x-3)$
 (c) Both $(x-1)$ and $(x-3)$ (d) Neither $(x-1)$ nor $(x-3)$
- 64.** Which one of the following is correct? ☑ 2015 II
 (a) $(x+2)$ is a factor of $x^4 - 6x^3 + 12x^2 - 24x + 32$
 (b) $(x+2)$ is a factor of $x^4 + 6x^3 - 12x^2 + 24x - 32$
 (c) $(x-2)$ is a factor of $x^4 - 6x^3 + 12x^2 - 24x + 32$
 (d) $(x-2)$ is a factor of $x^4 + 6x^3 - 12x^2 + 24x - 32$
- 65.** For what value of k $(x+1)$ is a factor of $x^3 + kx^2 - x + 2$? ☑ 2016 I
 (a) 4 (b) 3 (c) 1 (d) -2

- 66.** If the polynomial $x^6 + px^5 + qx^4 - x^2 - x - 3$ is divisible by $(x^4 - 1)$, then the value of $p^2 + q^2$ is ☑ 2016 I
 (a) 1 (b) 9 (c) 10 (d) 13
- 67.** If the linear factors of $ax^2 - (a^2 + 1)x + a$ are p and q , then $p + q$ is equal to ☑ 2016 I
 (a) $(x-1)(a+1)$ (b) $(x+1)(a+1)$
 (c) $(x-1)(a-1)$ (d) $(x+1)(a-1)$
- 68.** If $\sqrt{\frac{x}{y}} = \frac{10}{3} - \sqrt{\frac{y}{x}}$ and $x - y = 8$, then the value of xy is equal to ☑ 2016 I
 (a) 36 (b) 24 (c) 16 (d) 9

ANSWERS

1	b	2	b	3	c	4	a	5	a	6	a	7	b	8	a	9	d	10	a
11	d	12	c	13	b	14	b	15	c	16	b	17	d	18	d	19	d	20	b
21	b	22	c	23	d	24	a	25	c	26	c	27	d	28	b	29	a	30	b
31	b	32	d	33	c	34	b	35	c	36	a	37	b	38	c	39	c	40	d
41	d	42	c	43	c	44	c	45	a	46	d	47	c	48	d	49	d	50	b
51	c	52	c	53	d	54	d	55	c	56	b	57	c	58	a	59	a	60	c
61	a	62	b	63	c	64	c	65	d	66	c	67	a	68	d				

HINTS AND SOLUTIONS

1. (b) $x^2 + y^2 + z^2 - 2xy + 2yz - 2zx$
 $= (x - y - z)^2 = (5 - 3 - 2)^2 = 0$
 $[\because x = 5, y = 3 \text{ and } z = 2]$

2. (b) $x^2 - 2\sqrt{3}x + 3$
 $= x^2 - \sqrt{3}x - \sqrt{3}x + 3$
 $= x(x - \sqrt{3}) - \sqrt{3}(x - \sqrt{3})$
 $= (x - \sqrt{3})(x - \sqrt{3}) = (x - \sqrt{3})^2$

3. (c) $(a^4b^4 - 16c^4) = [(a^2b^2)^2 - (4c^2)^2]$
 $= [a^2b^2 + 4c^2][a^2b^2 - 4c^2]$
 $[\because a^2 - b^2 = (a - b)(a + b)]$
 $= [a^2b^2 + 4c^2][(ab)^2 - (2c)^2]$
 $= [a^2b^2 + 4c^2][(ab + 2c)(ab - 2c)]$

4. (a) $x^8 - y^8 = [(x^4)^2 - (y^4)^2]$
 $= [x^4 + y^4][x^4 - y^4]$
 $= [x^4 + y^4][(x^2)^2 - (y^2)^2]$
 $= (x^4 + y^4)(x^2 + y^2)(x^2 - y^2)$
 $= (x^4 + y^4)(x^2 + y^2)(x + y)(x - y)$

5. (a) $\left(a^2 + a + \frac{1}{4}\right) = a^2 + \frac{1}{2}a + \frac{1}{2}a + \frac{1}{4}$
 $= a\left(a + \frac{1}{2}\right) + \frac{1}{2}\left(a + \frac{1}{2}\right)$
 $= \left(a + \frac{1}{2}\right)\left(a + \frac{1}{2}\right) = \left(a + \frac{1}{2}\right)^2$

6. (a) $8 - 4x - 2x^3 + x^4$
 $= 4(2 - x) - x^3(2 - x)$
 $= (2 - x)(4 - x^3)$

7. (b) $(a^2 - b^2 - 4ac + 4c^2)$
 $= (a^2 - 4ac + 4c^2) - b^2$
 $[(a)^2 - 2(2c)(a) + (2c)^2] - b^2$
 $= (a - 2c)^2 - b^2$
 $= (a - 2c - b)(a - 2c + b)$

8. (a) $x^2 + 4y^2 + 4y - 4xy - 2x - 8$
 $= x^2 - 4xy + 4y^2 - 2x + 4y - 8$
 $= (x - 2y)^2 - 2(x - 2y) - 8$
 Let $x - 2y = z$

$\therefore z^2 - 2z - 8 = z^2 - 4z + 2z - 8$
 $= z(z - 4) + 2(z - 4)$
 $= (z + 2)(z - 4)$

Now, put $z = x - 2y$
 $\therefore (x - 2y + 2)$
 and $(x - 2y - 4)$ are required factors.

9. (d) $a^3 - \frac{1}{a^3} - 2a + \frac{2}{a}$
 $= a^3 - \left(\frac{1}{a}\right)^3 - 2\left(a - \frac{1}{a}\right)$
 $= \left(a - \frac{1}{a}\right)\left(a^2 + \frac{1}{a^2} + 1\right) - 2\left(a - \frac{1}{a}\right)$
 $= \left(a - \frac{1}{a}\right)\left(a^2 + \frac{1}{a^2} - 1\right)$

10. (a) $\left(\frac{1}{3}x^2 - 2x - 9\right) = \frac{(x^2 - 6x - 27)}{3}$
 $= \frac{1}{3}[x^2 - 9x + 3x - 27]$
 $= \frac{1}{3}[x(x - 9) + 3(x - 9)]$
 $= \frac{1}{3}(x + 3)(x - 9)$

$$\begin{aligned}
 11. (d) & 6\sqrt{3}x^2 - 47x + 5\sqrt{3} \\
 &= 6\sqrt{3}x^2 - 45x - 2x + 5\sqrt{3} \\
 &= 3\sqrt{3}x(2x - 5\sqrt{3}) - 1(2x - 5\sqrt{3}) \\
 &= (2x - 5\sqrt{3})(3\sqrt{3}x - 1)
 \end{aligned}$$

$$\begin{aligned}
 12. (c) & 8a^3 + 125b^3 - 64c^3 + 120abc \\
 &= (2a)^3 + (5b)^3 - (4c)^3 + 120abc \\
 &= (2a)^3 + (5b)^3 + (-4c)^3 - 3(2a)(5b)(-4c) \\
 &= (2a + 5b - 4c)(4a^2 + 25b^2 + 16c^2 - 10ab + 20bc + 8ac) \\
 &[\because a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)]
 \end{aligned}$$

$$\begin{aligned}
 13. (b) & \text{Since, } x^{1/3} + y^{1/3} + z^{1/3} = 0 \\
 \therefore & (x^{1/3})^3 + (y^{1/3})^3 + (z^{1/3})^3 - 3x^{1/3}y^{1/3}z^{1/3} = 0 \\
 \Rightarrow & x + y + z - 3(xyz)^{1/3} = 0 \\
 \Rightarrow & x + y + z = 3(xyz)^{1/3} \\
 \Rightarrow & (x + y + z)^3 = 27xyz
 \end{aligned}$$

$$\begin{aligned}
 14. (b) & \because \left(x + \frac{1}{x}\right)^3 = x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right) \\
 & (\sqrt{5})^3 = x^3 + \frac{1}{x^3} + 3(\sqrt{5}) \\
 & \left[\because x + \frac{1}{x} = \sqrt{5}\right] \\
 \Rightarrow & x^3 + \frac{1}{x^3} = 5\sqrt{5} - 3\sqrt{5} = 2\sqrt{5}
 \end{aligned}$$

$$\begin{aligned}
 15. (c) & \text{Given, } \left(x + \frac{1}{x}\right) = 6 \\
 \text{On squaring both sides, we get} \\
 & \left(x + \frac{1}{x}\right)^2 = 6^2 \\
 \Rightarrow & x^2 + \frac{1}{x^2} + 2 = 36 \\
 \Rightarrow & x^2 + \frac{1}{x^2} = 34
 \end{aligned}$$

$$\begin{aligned}
 16. (b) & \text{Given, } x - \frac{1}{x} = \frac{1}{3} \\
 \Rightarrow & 3x - \frac{3}{x} = 1 \\
 \text{On squaring both sides, we get} \\
 & \left(3x - \frac{3}{x}\right)^2 = (1)^2
 \end{aligned}$$

$$\begin{aligned}
 & 9x^2 + \frac{9}{x^2} - 2 \times 9 = 1 \\
 \Rightarrow & 9x^2 + \frac{9}{x^2} = 19
 \end{aligned}$$

$$\begin{aligned}
 17. (d) & \because (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca) \\
 \therefore & (6)^2 = 26 + 2(ab + bc + ca) \\
 \Rightarrow & 2(ab + bc + ca) = 10 \\
 \Rightarrow & ab + bc + ca = 5
 \end{aligned}$$

$$\begin{aligned}
 18. (d) & \text{Given, } x^4 + \frac{1}{x^4} = 322 \\
 \Rightarrow & \left(x^2 + \frac{1}{x^2}\right)^2 - 2 = 322 \\
 \Rightarrow & \left(x^2 + \frac{1}{x^2}\right)^2 = 324 = 18^2 \\
 \Rightarrow & x^2 + \frac{1}{x^2} = 18 \\
 \Rightarrow & \left(x - \frac{1}{x}\right)^2 + 2 = 18 \Rightarrow x - \frac{1}{x} = 4
 \end{aligned}$$

$$\begin{aligned}
 19. (d) & \text{Given, } a + b + c = 0 \\
 \Rightarrow & a^3 + b^3 + c^3 = 3abc \\
 \text{On dividing both sides by } abc, \text{ we get} \\
 \Rightarrow & \frac{a^3}{abc} + \frac{b^3}{abc} + \frac{c^3}{abc} = 3 \\
 \Rightarrow & \frac{a^2}{bc} + \frac{b^2}{ac} + \frac{c^2}{ab} = 3
 \end{aligned}$$

$$\begin{aligned}
 20. (b) & \because (a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ac \\
 & (10)^2 = (a^2 + b^2 + c^2) + 2(31) \\
 & a^2 + b^2 + c^2 = 100 - 62 \\
 \Rightarrow & a^2 + b^2 + c^2 = 38
 \end{aligned}$$

$$\begin{aligned}
 21. (b) & x^2 + \frac{1}{4x^2} + 1 - 2x - \frac{1}{x} \\
 &= \left(x^2 + \frac{1}{4x^2} + 1\right) - 2\left(x + \frac{1}{2x}\right) \\
 &= \left(x + \frac{1}{2x}\right)^2 - 2\left(x + \frac{1}{2x}\right) \\
 &= \left(x + \frac{1}{2x}\right)\left(x + \frac{1}{2x} - 2\right)
 \end{aligned}$$

$$\begin{aligned}
 22. (c) & (a^2 - b^2)(c^2 - d^2) - 4abcd \\
 &= a^2c^2 - a^2d^2 - b^2c^2 + b^2d^2 - 4abcd \\
 &= (a^2c^2 + b^2d^2 - 2abcd) - (b^2c^2 + a^2d^2 + 2abcd) \\
 &= (ac - bd)^2 - (bc + ad)^2 \\
 &= (ac - bd + bc + ad)(ac - bd - bc - ad)
 \end{aligned}$$

$$\begin{aligned}
 23. (d) & \because \left(a - \frac{1}{a}\right) = \sqrt{\left(a + \frac{1}{a}\right)^2 - 4} \\
 &= \sqrt{\left(\frac{17}{4}\right)^2 - 4} \\
 & \left[\because \text{given } \left(a + \frac{1}{a}\right) = \frac{17}{4}\right] \\
 &= \sqrt{\frac{289 - 64}{16}} = \sqrt{\frac{225}{16}} = \frac{15}{4}
 \end{aligned}$$

$$\begin{aligned}
 24. (a) & x^4 + 4y^4 \\
 &= x^4 + 4y^4 + 4x^2y^2 - 4x^2y^2 \\
 &= (x^2 + 2y^2)^2 - (2xy)^2 \\
 &= (x^2 + 2y^2 - 2xy)(x^2 + 2y^2 + 2xy)
 \end{aligned}$$

From above it is clear that $x^4 + 4y^4$ is divisible by $x^2 + 2y^2 + 2xy$

$$\begin{aligned}
 25. (c) & \text{Given, } x(x + y + z) = 9, \\
 & y(x + y + z) = 16 \\
 & \text{and } z(x + y + z) = 144 \\
 \text{On adding all three equations, we get} \\
 & x(x + y + z) + y(x + y + z) + z(x + y + z) \\
 &= 9 + 16 + 144 \\
 \Rightarrow & (x + y + z)(x + y + z) = 9 + 16 + 144 \\
 \Rightarrow & (x + y + z)^2 = 169 \\
 \Rightarrow & x + y + z = 13 \\
 \therefore & x(x + y + z) = 9 \\
 \Rightarrow & x(13) = 9 \Rightarrow x = \frac{9}{13}
 \end{aligned}$$

$$\begin{aligned}
 26. (c) & \text{Given, } x = (b - c)(a - d), \\
 & y = (c - a)(b - d) \\
 \text{and } & z = (a - b)(c - d) \\
 \therefore & x + y + z = (b - c)(a - d) + (c - a)(b - d) + (a - b)(c - d) \\
 &= 0 \\
 \text{Hence, } & x^3 + y^3 + z^3 = 3xyz
 \end{aligned}$$

$$\begin{aligned}
 27. (d) & \text{Here, } px^3 + x^2 - 2x - q \text{ is divisible by } (x - 1) \text{ and } (x + 1). \\
 \therefore & p(1)^3 + (1)^2 - 2(1) - q = 0 \quad \dots(i) \\
 \Rightarrow & p - q = 1 \\
 \text{and } & p(-1)^3 + (-1)^2 - 2(-1) - q = 0 \quad \dots(ii) \\
 \Rightarrow & -p + q = 3
 \end{aligned}$$

On solving Eqs. (i) and (ii), we get

$$\begin{aligned}
 & p = 2 \text{ and } q = 1 \\
 28. (b) & \text{Let } f(x) = px^3 + 3x^2 - 3 \\
 \text{and } & g(x) = 2x^3 - 5x + p. \\
 \text{On dividing } f(x) \text{ by } (x - 4), \text{ remainder is} \\
 & f(4) = p(4)^3 + 3(4)^2 - 3 \\
 &= 64p + 48 - 3 = 64p + 45 \\
 \text{Similarly } & g(4) = 2(4)^3 - 5(4) + p \\
 &= 128 - 20 + p = 108 + p \\
 \text{But } & f(4) = g(4) \quad [\text{given}] \\
 \therefore & 64p + 45 = 108 + p \\
 \Rightarrow & 63p = 63 \Rightarrow p = 1
 \end{aligned}$$

$$\begin{aligned}
 29. (a) & ax - b = 0 \Rightarrow x = b/a \\
 \text{By remainder theorem, if } f(x) \text{ is divisible by } ax - b, \text{ the remainder is } f(b/a).
 \end{aligned}$$

30. (b) Factor theorem is a special case of remainder theorem.

$$\begin{aligned}
 31. (b) & x(y - z)(y + z) + y(z - x)(z + x) \\
 &= x(y^2 - z^2) + y(z^2 - x^2) + z(x^2 - y^2) \\
 &= x(y^2 - z^2) + yz^2 - yx^2 + zx^2 - zy^2 \\
 &= (y - z)(xy + xz - x^2 - yz) \\
 &= (y - z)[y(x - z) + x(z - x)] \\
 &= (y - z)(z - x)(x - y) \\
 &= (x - y)(x - z)(z - y)
 \end{aligned}$$

32. (d) Let $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$

Since, $(x-1)$ is a factor of $f(x)$.

Put $x = 1$ in $f(x)$, then

$$f(1) = a_0 + a_1 + a_2 + \dots + a_n$$

$$\Rightarrow 1 = a_0 + a_1 + a_2 + \dots + a_n$$

$$\therefore 1 - a_0 - a_2 - \dots = a_1 + a_3 + \dots$$

33. (c) Given, $(u)^3 + (-2v)^3 + (-3w)^3 = 3 \times (-2)(-3)uvw$

$$\therefore u + (-2v) + (-3w) = 0$$

$$[\because a^3 + b^3 + c^3 = 3abc, \text{ then } a + b + c = 0]$$

$$\Rightarrow u - 2v - 3w = 0$$

$$\therefore u - 2v = 3w$$

34. (b) We know that, $(a+b)(a-b)$

$$= a^2 - b^2$$

So, $[(a^{1/8} + a^{-1/8})(a^{1/8} - a^{-1/8})$

$$(a^{1/4} + a^{-1/4})(a^{1/2} + a^{-1/2})]$$

$$= [(a^{1/8})^2 - (a^{-1/8})^2](a^{1/4} + a^{-1/4})]$$

$$= (a^{1/4} - a^{-1/4})(a^{1/4} + a^{-1/4})$$

$$(a^{1/2} + a^{-1/2})$$

$$= [a^{1/2} - a^{-1/2}][a^{1/2} + a^{-1/2}]$$

$$= (a - a^{-1})$$

35. (c) $f(x) = x^4 - 2x^3 + 3x^2 - ax + b$

Put $x = 1$ as $f(1) = 5$, $f(-1) = 19$

$$f(1) = 1^4 - 2(1)^3 + 3(1)^2 - a + b$$

$$\Rightarrow 1 - 2 + 3 - a + b = 5$$

$$\Rightarrow -a + b = 3 \quad \dots(i)$$

and $f(-1) = (-1)^4 - 2(-1)^3 + 3$

$$(-1)^2 - a(-1) + b$$

$$\Rightarrow 1 + 2 + 3 + a + b = 19$$

$$\Rightarrow a + b = 13 \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get

$$a = 5, b = 8$$

36. (a) Let $f(x) = ax^2 + bx + c$

Given $f(0) = 4$

$$\Rightarrow 0 + 0 + c = 4 \Rightarrow a - b + 4 = 4 \Rightarrow c = 4$$

On dividing $f(x)$ by $x + 1$, remainder is 4

$$\therefore f(-1) = 4$$

$$\Rightarrow a - b + c = 4 \Rightarrow a - b + 4 = 4$$

$$\Rightarrow a - b = 0 \quad \dots(i)$$

On dividing $f(x)$ by $x + 2$, remainder is 6

$$\therefore f(-2) = 6$$

$$\Rightarrow 4a - 2b + c = 6$$

$$\Rightarrow 4a - 2b = 2 \quad \dots(ii)$$

$$\Rightarrow a - b + 4 = 4$$

On solving Eq. (i) and eq. (ii), we get

$$a = 1, b = 1$$

Hence,

$$a = 1, b = 1, c = 4.$$

37. (b) $x^{3/2} - xy^{1/2} + x^{1/2}y - y^{3/2}$

$$= x(x^{1/2} - y^{1/2}) + y(x^{1/2} - y^{1/2})$$

$$= (x^{1/2} - y^{1/2})(x + y)$$

$\Rightarrow$ Quotient

$$= \frac{(x^{1/2} - y^{1/2})(x + y)}{(x^{1/2} - y^{1/2})} = x + y$$

38. (c) Let $f(x) = 9x^2 + 3px + 6q$

Given, $f(-1/3) = -3/4$

$$\Rightarrow 9\left(-\frac{1}{3}\right)^2 + 3p\left(-\frac{1}{3}\right) + 6q = -3/4$$

$$\Rightarrow 1 - p + 6q = -3/4$$

$$\Rightarrow 24q - 4p + 7 = 0 \quad \dots(i)$$

Let $g(x) = qx^2 + 4px + 7$

Since, $(x + 1)$ is a factor of $g(x)$

$$\therefore g(-1) = 0 \Rightarrow q - 4p + 7 = 0 \dots(ii)$$

On solving Eq. (i) and Eq. (ii), we get

$$q = 0 \text{ and } p = 7/4$$

39. (c) I. $(a+b)(b+c)(c+a)$

$$= (1-c)(1-a)(1-b)$$

$$[\because a + b + c = 1]$$

$$= (1-c)[1 - b - a + ab]$$

$$= 1 - b - a + ab - c + bc + ac - abc$$

$$= 1 - (a + b + c) + ab + bc + ac - abc$$

$$= ab + bc + ac - abc$$

II. $a^2 + b^2 - c^2 + 2ab$

$$= a^2 + b^2 + 2ab - c^2$$

$$= (a + b)^2 - c^2$$

$$= (a + b + c)(a + b - c) = a + b - c$$

$$[\because a + b + c = 1]$$

So, both statements are correct.

40. (d) We have, $x + \frac{1}{x} = 4$

I. $\left(x + \frac{1}{x}\right)^3 = x^3 + \frac{1}{x^3} + 3(x)$

$$\times \left(\frac{1}{x}\right) \left(x + \frac{1}{x}\right)$$

$$(4)^3 = x^3 + \frac{1}{x^3} + 3(4)$$

$$64 - 12 = x^3 + \frac{1}{x^3} \Rightarrow x^3 + \frac{1}{x^3} = 52$$

II. $x + \frac{1}{x} = 4 \Rightarrow x^2 - 4x + 1 = 0$

$$x = \frac{4 \pm \sqrt{16 - 4 \times 1 \times 1}}{2 \times 1}$$

$$= \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}$$

Hence, both statements are correct.

41. (d) Let $f(x) = 4x^4 - (k-1)x^3 + kx^2 - 6x + 1 \quad \dots(i)$

Since, $(2x - 1)$ is a factor of $f(x)$

$$\therefore f\left(\frac{1}{2}\right) = 0$$

$$\Rightarrow 4 \times \left(\frac{1}{2}\right)^4 - (k-1) \times \left(\frac{1}{2}\right)^3 + k \times \left(\frac{1}{2}\right)^2 - 6 \times \frac{1}{2} + 1 = 0$$

$$\Rightarrow \frac{1}{4} - \frac{(k-1)}{8} + \frac{k}{4} - 2 = 0$$

$$\Rightarrow \frac{k}{4} - \frac{(k-1)}{8} = 2 - \frac{1}{4}$$

$$\Rightarrow \frac{2k - (k-1)}{8} = \frac{8-1}{4} = \frac{7}{4}$$

$$\Rightarrow \frac{8k - 4k + 4}{8} = 56$$

$$\Rightarrow 4k = 52 \Rightarrow k = 13$$

Hence, the value of k is 13.

42. (c) Given, $x^4 + xy^3 + x^3y + xz^3 + y^4 + yz^3$

$$= (x^4 + xy^3 + xz^3) + (x^3y + y^4 + yz^3)$$

$$= x(x^3 + y^3 + z^3) + y(x^3 + y^3 + z^3)$$

$$= (x + y)(x^3 + y^3 + z^3)$$

Hence, given polynomial is divisible by both

$$(x + y) \text{ and } (x^3 + y^3 + z^3).$$

43. (c) $x^2 + 2 \mid x^3 + 5x^2 + 10k(x + 5)$

$$\frac{x^3 + 2x}{5x^2 - 2x + 10k}$$

$$\frac{5x^2 + 10}{-2x - 10 + 10k} = \text{Remainder}$$

$$\frac{-2x - 10 + 10k}{-2x - 10 + 10k} = \text{Remainder}$$

But given that, remainder = $-2x$

$$\therefore -2x - 10 + 10k = -2x$$

$$\Rightarrow -10 + 10k = 0$$

$$\Rightarrow 10k = 10 \Rightarrow k = 1$$

44. (c) Given that, $x + \frac{1}{x} = a \quad \dots(i)$

Then, $x^3 + x^2 + \frac{1}{x^3} + \frac{1}{x^2}$

$$= \left(x^3 + \frac{1}{x^3}\right) + \left(x^2 + \frac{1}{x^2}\right)$$

$$= \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right) + \left(x + \frac{1}{x}\right)^2 - 2$$

$$= a^3 - 3a + a^2 - 2 = a^3 + a^2 - 3a - 2$$

45. (a) Given, $f(x) = x^3 + x^2 - x - 1$

and $g(x) = x^3 - x^2 + x - 1$

$$p(x) = f(x) \cdot g(x)$$

$$= (x^3 + x^2 - x - 1)(x^3 - x^2 + x - 1)$$

$$= [x^2(x+1) - 1(x+1)]$$

$$[x^2(x-1) + 1(x-1)]$$

$$= [(x+1)(x^2-1)][(x-1)(x^2+1)]$$

$$= (x+1)(x+1)(x-1)(x-1)(x^2+1)$$

$$= (x+1)^2(x-1)^2(x^2+1)$$

46. (d) Put the value of all options in
 $4k^2 + 12k + k$
 when, $k = 9$,
 then, $4(9)^2 + 12(9) + 9 = 441 = (21)^2$

47. (c) Given expression
 $= 5px - 10qy + 2rpx - 4qry$
 $= (5px + 2rpx) - (10qy + 4qry)$
 $= px(5 + 2r) - 2qy(5 + 2r)$
 $= (5 + 2r)(px - 2qy)$
 So, factors of given expression are both
 $(5 + 2r)$ and $(px - 2qy)$.

48. (d) Given, $(a + 1)^4 - a^4$
 $= \{(a + 1)^2 - a^2\} \{(a + 1)^2 + a^2\}$
 $= \{(a + 1) + a\} \{(a + 1) - a\}$
 $\quad \quad \quad \{a^2 + 1 + 2a + a^2\}$
 $= (2a + 1)(1)(2a^2 + 2a + 1)$
 $= (2a + 1)(2a^2 + 2a + 1)$
 So, $(a + 1)^4 - a^4$ is divisible by
 $(2a^2 + 2a + 1)$

49. (d) Let $f(x) = x^4 - 7x^3 + 5x^2 - 6x + 81$
 $f(3) = (3)^4 - 7(3)^3 + 5(3)^2 - 6(3) + 81$
 $= 81 - 189 + 45 - 18 + 81 = 0$
 $\therefore (x - 3)$ is a factor of $f(x)$

50. (b) Let $f(x) = x^3 + 3x^2 + 4x + k$
 $\therefore (x + 5)$ is a factor of $f(x)$
 $\therefore f(-5) = 0$
 $\Rightarrow (-5)^3 + 3(-5)^2 + 4(-5) + k = 0$
 $\Rightarrow -125 + 75 - 20 + k = 0$
 $\Rightarrow -70 + k = 0 \Rightarrow k = 70$
 Hence, the value of k is 70.

51. (c) Let $p(x)$ be added to obtain x^3 .
 Then, $(1 - x)(1 + x^2) + p(x) = x^3$
 $\Rightarrow 1 + x^2 - x - x^3 + p(x) = x^3$
 $\Rightarrow p(x) = x^3 - 1 - x^2 + x + x^3$
 $\therefore p(x) = 2x^3 - x^2 + x - 1$

So, $2x^3 - x^2 + x - 1$ is added to
 $(1 - x)(1 + x^2)$ to obtain x^3 .

52. (c) Let $f(x) = x(y^2 - z^2) + y(z^2 - x^2)$
 $\quad \quad \quad + z(x^2 - y^2)$

If it is divisible by $(y - z)$, then $y - z$ is
 a factor of $f(x)$.

$$\therefore y - z = 0 \Rightarrow y = z$$

On putting $y = z$, we get

$$x(z^2 - z^2) + z(z^2 - x^2) + z(x^2 - z^2)$$

$$= z^3 - zx^2 + zx^2 - z^3 = 0$$

Hence, $y - z$ is a factor, so it is divisible
 by $(y - z)$.

On putting $z = x$, we get

$$x(y^2 - x^2) + y(x^2 - x^2)$$

$$\quad \quad \quad + x(x^2 - y^2)$$

$$= xy^2 - x^3 + x^3 - xy^2 = 0$$

So, $(z - x)$ is also a factor, so it is also
 divisible by $(z - x)$.

53. (d) Put $x = -3$ in $p(x)$, we get

$$\text{Let } p(x) = x^3 + 2x^2 + 3x + 8$$

$$p(-3) = (-3)^3 + 2(-3)^2 + 3(-3) + 8$$

$$= -27 + 18 - 9 + 8$$

$$= -10 \neq 0$$

So, $(x + 3)$ is not the factor of
 $x^3 + 2x^2 + 3x + 8$.

Similarly, put $x = 2$ in $p(x)$, we get

$$p(2) = (2)^3 + 2(2)^2 + 3(2) + 8$$

$$= 30 \neq 0 \text{ So, } x - 2 \text{ is also not the factor}$$

$$\text{of } x^3 + 2x^2 + 3x + 8.$$

54. (d) Let $f(x) = x^3 + 6x^2 + 11x + 6$

Put $x = -1$ in $f(x)$, we get

$$f(-1) = (-1)^3 + 6(-1)^2 + 11(-1) + 6$$

$$= -1 + 6 - 11 + 6$$

$$= -12 + 12 = 0$$

Hence, $(x + 1)$ is a factor of $f(x)$.

$$\therefore f(x) = (x + 1)(x^2 + 5x + 6)$$

$$= (x + 1)(x^2 + 3x + 2x + 6)$$

$$= (x + 1)(x + 2)(x + 3)$$

Hence, $(x + 1)$, $(x + 2)$ and $(x + 3)$ are
 the factors of $f(x)$.

55. (c) $x(x + a)(x + 2a)(x + 3a)$

$$= (x + a)(x + 2a)x(x + 3a)$$

$$= (x^2 + 3ax + 2a^2)(x^2 + 3ax)$$

$$= (x^2 + 3ax + a^2 + a^2)$$

$$\quad \quad \quad (x^2 + 3ax + a^2 - a^2)$$

$$= (x^2 + 3ax + a^2)^2 - a^4$$

So, a^4 must be added to
 $x(x + a)(x + 2a)(x + 3a)$ to make it a
 perfect square.

56. (b) Given, $\left(x^2 + \frac{1}{x^2}\right) = \frac{17}{4}$

$$\Rightarrow x^2 + \frac{1}{x^2} - 2 + 2 = \frac{17}{4}$$

$$\Rightarrow \left(x - \frac{1}{x}\right)^2 + 2 = \frac{17}{4}$$

$$\Rightarrow \left(x - \frac{1}{x}\right)^2 = \frac{17}{4} - 2$$

$$\Rightarrow \left(x - \frac{1}{x}\right)^2 = \frac{9}{4}$$

$$\Rightarrow \left(x - \frac{1}{x}\right) = \frac{3}{2} \quad \dots(i)$$

On cubing both sides, we get

$$\left(x - \frac{1}{x}\right)^3 = \left(\frac{3}{2}\right)^3$$

$$\Rightarrow x^3 - \frac{1}{x^3} - 3 \times \frac{1}{x} \cdot x \left(x - \frac{1}{x}\right) = \frac{27}{8}$$

$$\Rightarrow x^3 - \frac{1}{x^3} = \frac{27}{8} + 3 \times \left(\frac{3}{2}\right)$$

[From Eq. (i)]

$$\Rightarrow x^3 - \frac{1}{x^3} = \frac{27}{8} + \frac{9}{2}$$

$$\therefore \left(x^3 - \frac{1}{x^3}\right) = \frac{63}{8}$$

57. (c) Let $f(x) = x^5 - 5x^2 + 125$

Now, put $x = -5$,

$$\therefore \text{Required remainder} = f(-5)$$

$$= (-5)^5 - 5(-5)^2 + 125$$

$$= -3125 - 125 + 125 = -3125$$

58. (a) Given, $ax + by - 2 = 0$

$$\Rightarrow ax + by = 2$$

Squaring on both sides, we get

$$(ax + by)^2 = (2)^2$$

$$\Rightarrow a^2x^2 + b^2y^2 + 2axy = 4$$

$$\Rightarrow a^2x^2 + b^2y^2 + 2 = 4 \quad [\because axy = 1]$$

$$\Rightarrow a^2x^2 + b^2y^2 = 2$$

$$\Rightarrow a^2x^2 = 1 \text{ and } b^2y^2 = 1$$

$$\Rightarrow ax = 1 \text{ and } by = 1$$

$$\Rightarrow x = \frac{1}{a} \text{ and } y = \frac{1}{b}$$

$$\therefore a^2x + b^2y = a^2 \cdot \frac{1}{a} + b^2 \cdot \frac{1}{b} = a + b$$

59. (a) Given, $x + k$ is the common factor of
 $x^2 + ax + b$ and $x^2 + cx + d$. Then, put
 $x = -k$ in given polynomial.

$$\therefore k^2 - ka + b = 0$$

$$\text{and } k^2 - kc + d = 0$$

$$\text{Then, } k^2 - ka + b = k^2 - kc + d$$

$$\Rightarrow k(c - a) = d - b \Rightarrow k = \frac{d - b}{c - a}$$

60. (c) I. We have, $3abc + b^3 + c^3 - a^3$

$$= -(a^3 - b^3 - c^3 - 3abc)$$

$$= -[a^3 + (-b)^3 + (-c)^3 - 3(a)(-b)(-c)]$$

$$= -(a - b - c)(a^2 + b^2 + c^2 + ab - bc + ac)$$

So, $(a - b - c)$ is a factor of

$$3abc + b^3 + c^3 - a^3.$$

Hence, statement I is correct.

II. Now, $3bc + b^3 + c^3 - 1$

$$= b^3 + c^3 - (1)^3 - 3b(-1)(-1)$$

$$= (b + c - 1)[b^2 + c^2 + 1^2 - bc + c + b]$$

So, $(b + c - 1)$ is a factor of

$$3bc + b^3 + c^3 - 1.$$

Hence, statement II is also correct.